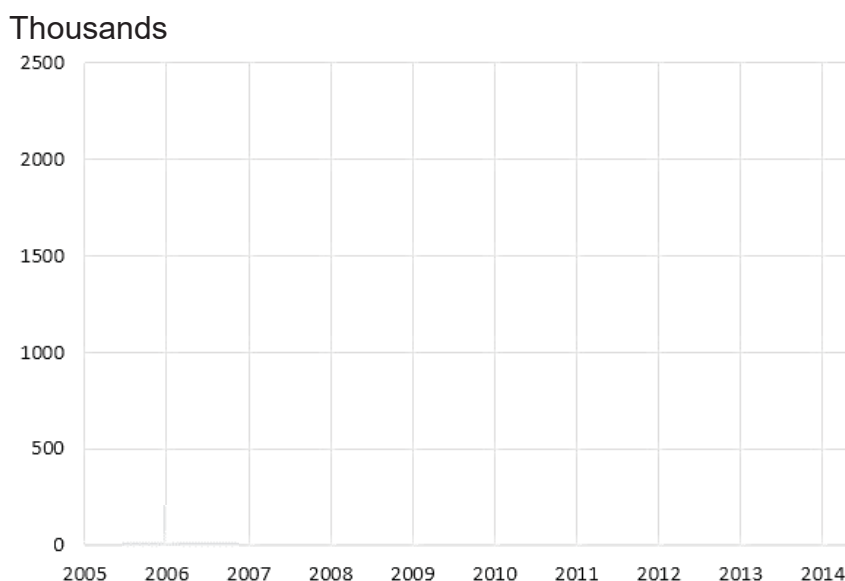


- (c) Draw a new graph using the information from the current graph and the axes below.

Answer

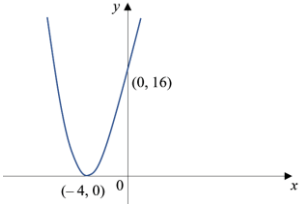
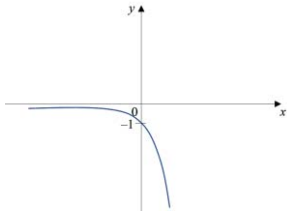
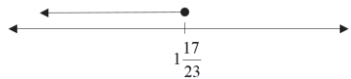
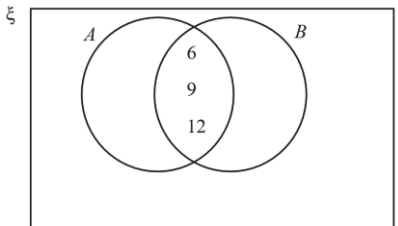
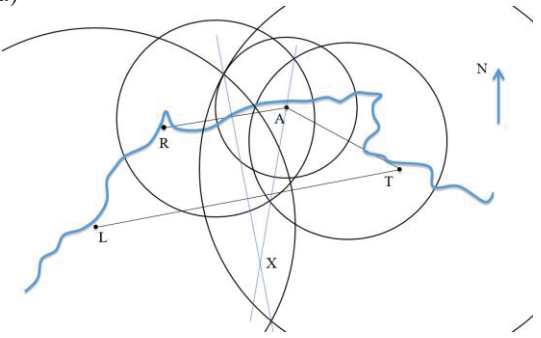
Cancer Screening and Abortion Chart



[1]

END OF PAPER

1	(a) 150.0401 (correct to 4 d.p.) (b) 150.0 (correct to 4 s.f.)	2	2.83×10^2 (3 s.f.)
3	(a) 10.3% (to 3 s.f.) (b) \$180	4	$3^{3x+2} - 9^{\frac{3}{2}x} + (27)^{x+1} = 3^{3x}(35)$ Since 35 is a multiple of 5, hence $3^{3x+2} - 9^{\frac{3}{2}x} + \left(\frac{1}{27}\right)^{-x-1}$ is divisible by 5 for all positive integer values.
5	(a) Smallest possible of the third number is 132 (i.e. $2^2 \times 3 \times 11$) (b) (i) $6468 = 2^2 \times 3 \times 7^2 \times 11$ (ii) $3 \times 11 = 33$		
6	(a) 2 hours (b) 48		
7	(a) 2.4 km (b) 6.25 cm		
8	(a) 2.5 m/s (b) 870 km (3 s.f.) or $869\frac{13}{23}$	9	$\frac{12x+5}{2(3x-1)(3x+1)}$
10	(a) $(b+1)(2a-1)(2a+1)$ (b) $x = \sqrt[3]{\frac{3g-f}{2}}$	11	(a) $(3x+4)(2x-5)$ (b) $y = -\frac{1}{3}$ or $y = \frac{7}{2}$

12	(a) (i) $y = (x + 4)^2$ (ii)  (b) $x = 0$ (c) $(-2, 0)$	13	(a)(i) $n = -1$ or $n = -3$ or Any negative odd number (ii) $k < 0$ (b) 
14	(b) $p = 5$ (c) 1	15	(a) $x \leq 1\frac{17}{23}$  (b) 0, 1
16	$n = 5$	18	(a) 4.5 km (b) 31 min (c) 0.36 km/min (2 d.p.)
17	$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$ $A = \{4, 6, 8, 9, 10, 12\}$ $B = \{3, 6, 9, 12\}$ $C = \{6, 12\}$ (a)(i) $A' = \{1, 2, 3, 5, 7, 11, 13\}$ or 1, 2, 3, 5, 7, 11, 13 (ii) $A' \cap C = \{ \}$ or $\emptyset$ or null/empty set or no element (b)  (c) $\{ \}, \{6\}, \{12\}$	19	(a) $\angle QRP = 90^\circ - 2x^\circ$ (supplementary angles)
21	(a) $\triangle CDE \equiv \triangle CBF$ (SSS) (b) 4 : 9 (c) 6cm	20	(a)  (b) $AX = 4.5 \times 200 = 900$ km *please note that you may not get the answer as 900km because the size of paper is being shrunk from A4 size to A5 size.
22	(a) 18 cm (b) $-\frac{3}{5}$		
23	(a) 133974.75 persons per year (b) The scale on the vertical axis is not defined.		Cancer Screening and Abortion Chart 